A UNIQUE PRIME OVER P PLUS GOING UP (OR DOWN) PINS DOWN THE LOCALISATION AT P

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Going up and going down as lifting properties for chains of primes. Used in the two branches of the counted proof (the diagram chase in lines 10035-10058). Carries a chain of primes in the base to a chain in the source, which is what forces the primes above p to be unique.
- Primes of a localisation correspond to primes below the localised prime. Used at the start of the counted proof (lines 10031-10034). Describes the points of the localisation as the primes whose contraction lies below p.

Lemma 0.1. *Let $R \to S$ be a ring map. Let $\mathfrak{p} \subset R$ be a prime. Assume that*

(1) *there exists a unique prime $\mathfrak{q} \subset S$ lying over $\mathfrak{p}$, and*
(2) *either*
   (a) *going up holds for $R \to S$, or*
   (b) *going down holds for $R \to S$ and there is at most one prime of S above every prime of R.*

Then $S_{\mathfrak{p}} = S_{\mathfrak{q}}$.

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Proof. Consider any prime $\mathfrak{q}' \subset S$ which corresponds to a point of $\mathrm{Spec}(S_\mathfrak{p})$. This means that $\mathfrak{p}' = R \cap \mathfrak{q}'$ is contained in $\mathfrak{p}$. Here is a picture

$$\begin{array}{ccccc} \mathfrak{q}' & \text{---} & ? & \text{---} & S \\ | & & | & & | \\ \mathfrak{p}' & \text{---} & \mathfrak{p} & \text{---} & R \end{array}$$

Assume (1) and (2)(a). By going up there exists a prime $\mathfrak{q}'' \subset S$ with $\mathfrak{q}' \subset \mathfrak{q}''$ and $\mathfrak{q}''$ lying over $\mathfrak{p}$. By the uniqueness of $\mathfrak{q}$ we conclude that $\mathfrak{q}'' = \mathfrak{q}$. In other words $\mathfrak{q}'$ defines a point of $\mathrm{Spec}(S_\mathfrak{q})$.

Assume (1) and (2)(b). By going down there exists a prime $\mathfrak{q}'' \subset \mathfrak{q}$ lying over $\mathfrak{p}'$. By the uniqueness of primes lying over $\mathfrak{p}'$ we see that $\mathfrak{q}' = \mathfrak{q}''$. In other words $\mathfrak{q}'$ defines a point of $\mathrm{Spec}(S_\mathfrak{q})$.

In both cases we conclude that the map $\mathrm{Spec}(S_\mathfrak{q}) \to \mathrm{Spec}(S_\mathfrak{p})$ is bijective. Clearly this means all the elements of $S - \mathfrak{q}$ are all invertible in $S_\mathfrak{p}$, in other words $S_\mathfrak{p} = S_\mathfrak{q}$. $\square$